



Idea to Production-Ready Agent in Seconds on AI-native Runtime



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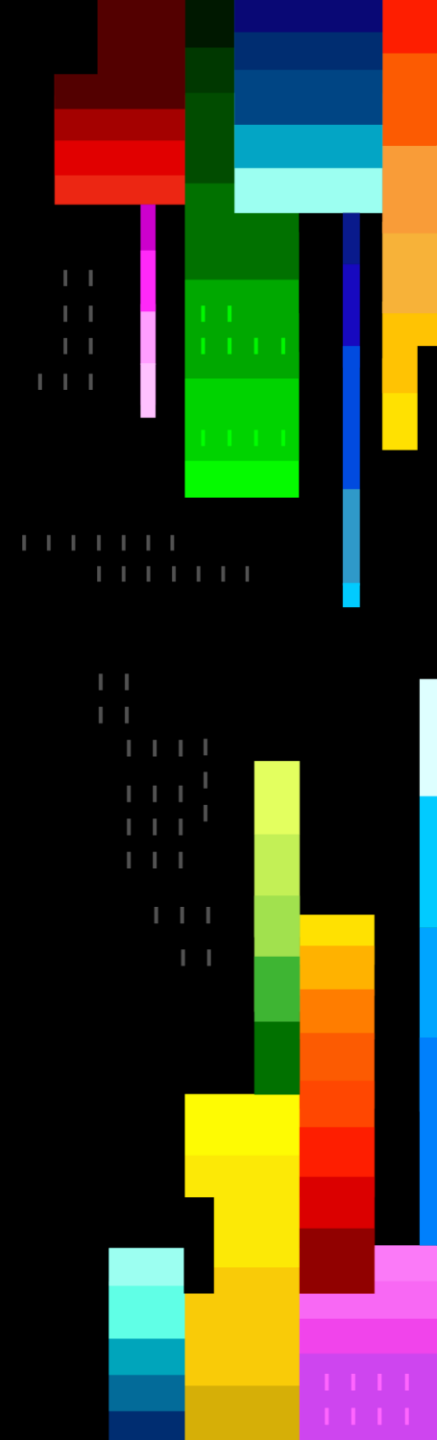
Simon Jakesch

Principal Product Manager, Azure Container Apps



Gopi Prashanth

Chief AI & Agents Office, Auger



Agenda



Why agents break between demo and production



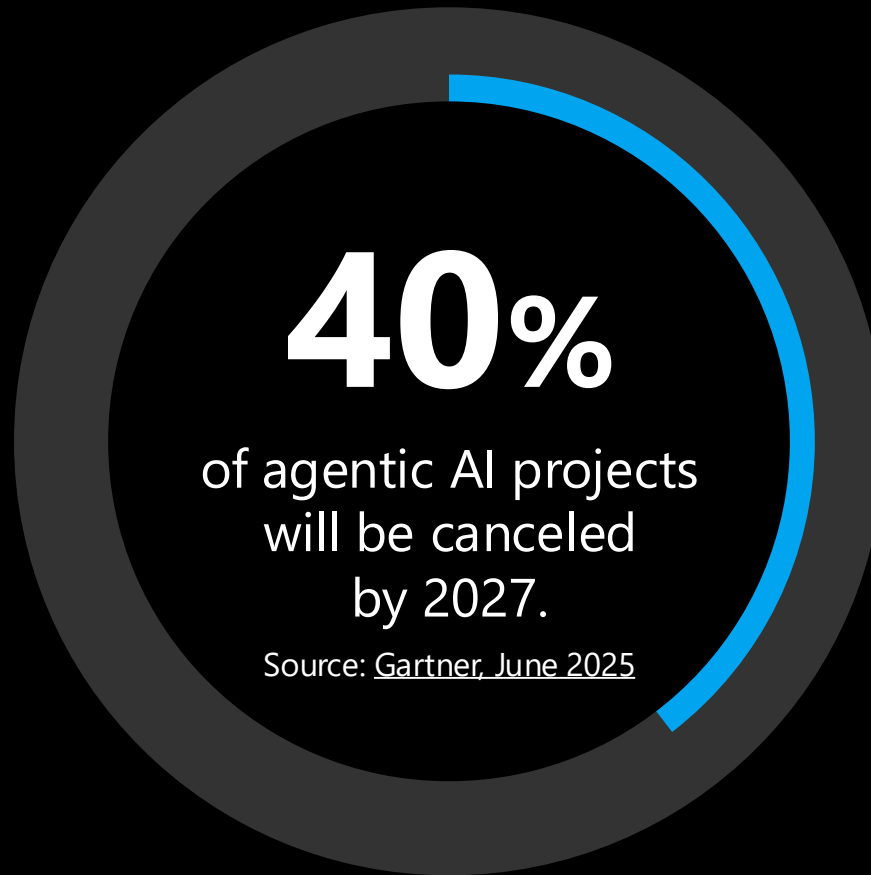
Live demo: Code to production-ready agent in seconds



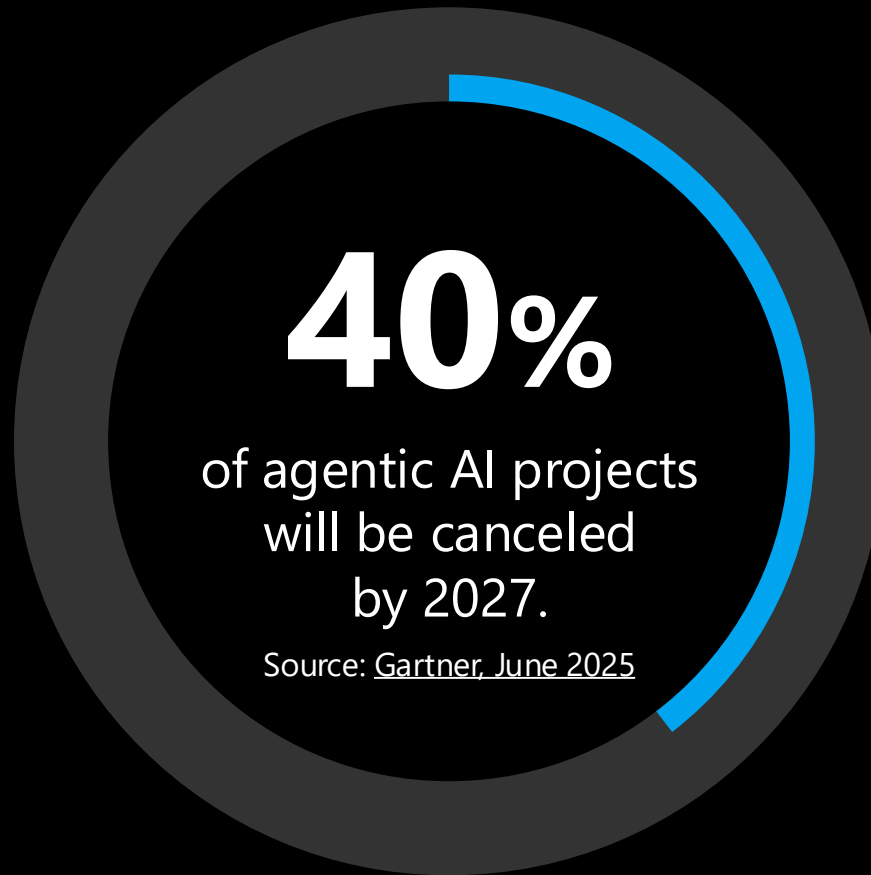
How Azure Container Apps delivers the agent runtime



Customer spotlight: Auger's autonomous supply chain on Azure



Not because the models can't reason. Because the runtime can't keep up.



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Five Places The Runtime Breaks



Budgets
that burn
unattended



Untrusted
code on a
developer's
laptop



Cold starts that
throttle the
loop



Workspaces
that die on
every restart



Tooling stitched
together by
hand

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Five Things The Runtime Has To Do



**Fast startup
and resume**

Agents remain
responsive and
reliable



**Execute agent
tool calling**

And untrusted
code, safely



**Persist and
restore state**

Support
long-running agent
workflows



**Strong isolation
per agent task**

Prevent cross-task
data leakage



**Secure by
default**

Enforced at the
runtime boundary

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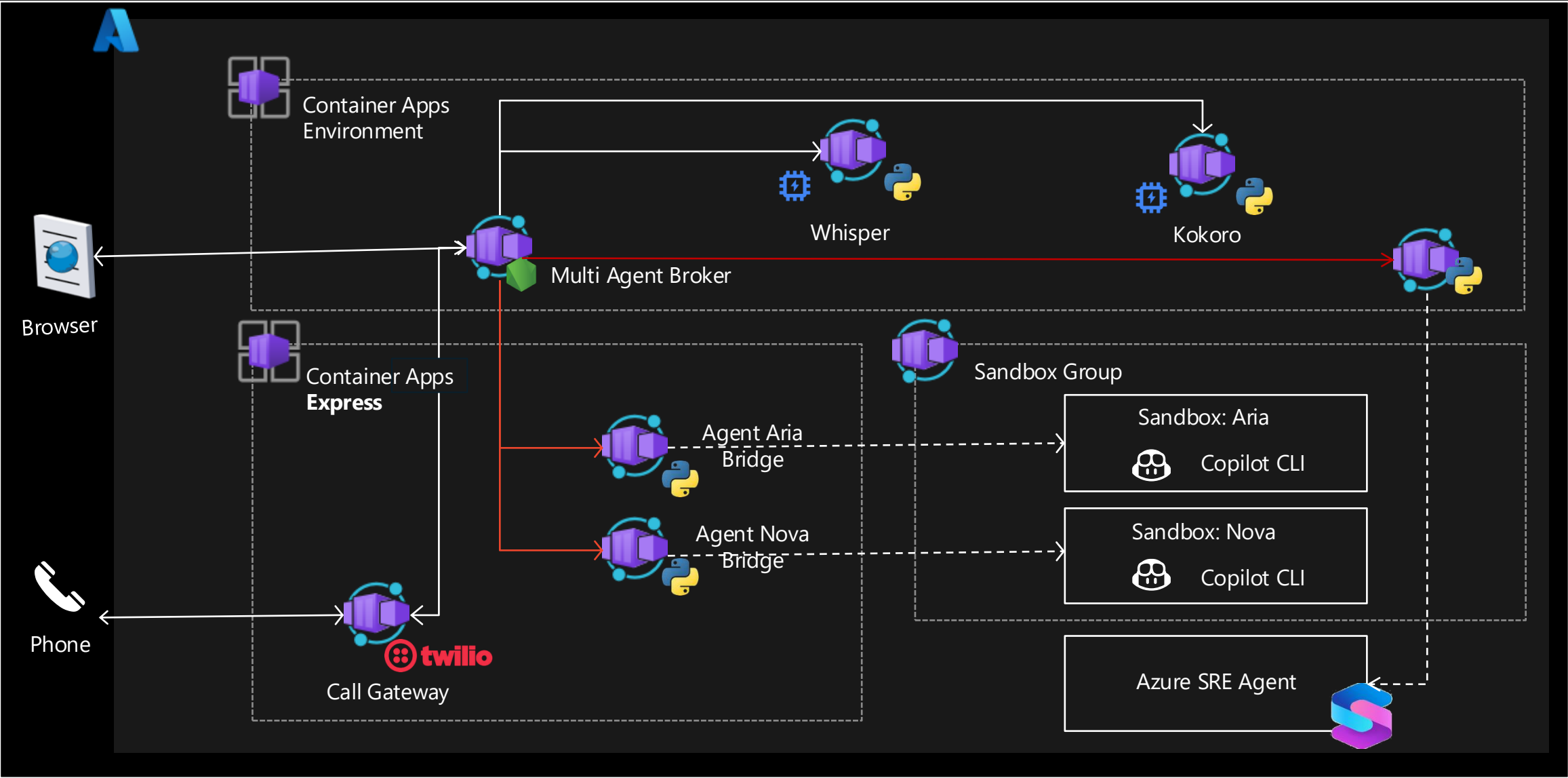
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Demo Scenario



aka.ms/aca/build2026-brk221



How Container Apps Cleared Every Runtime Hurdle

aka.ms/aca/build2026-brk221



Summary

One platform for apps, MCP tools, and ephemeral compute

Sub-second cold start	Express & Sandbox prewarmed pools
Strong per-task isolation	Code execution in Sandbox
Persistent agent workspace	Sandbox snapshots
Massive burst	Auto scale-out to thousands of concurrent sandboxes
Cost effective	Scale to zero when idle
GPU on demand	Serverless GPUs for inferencing
Zero infra decisions	Opinionated defaults with Express

One platform for apps, MCP tools, and ephemeral compute

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One platform for apps, MCP tools, and ephemeral compute

Introducing in Public Preview

Azure Container Apps Sandboxes

Fast, isolated and stateful compute infrastructure on demand

Execute Securely by Default

Sandbox isolation for any
untrusted workload

Resume Instantly

Preserve every interaction with trusted
enterprise controls

Burst to hyperscale

Spin up in sub-second time, scale from zero to
thousands, pay nothing when idle

Foundation layer for



aka.ms/aca/sandboxes

“

With Azure Container Apps sandboxes, SitecoreAI can safely enable agents to take real action. The combination of multi-tenant isolation, rapid scale-out, and full automation allows Sitecore to run long-lived, autonomous agents that securely execute code, manage workflows, and interact with enterprise systems within secure, governed environments.

With this foundation, we can build agents that do real work – assembling content, personalizing experiences, and optimizing campaigns in production. Agents that operate continuously, learn from results, and improve over time, so our customers get better outcomes without giving up control.”

Mo Cherif
VP of AI and Innovation, Sitecore



“

EdChat is an education AI platform where students learn safely with AI through coding, data analysis, and exploration. Azure Container Apps sandboxes would add built-in multi-tenant isolation and scale-to-zero, so each student could run in a cost-efficient, isolated environment with strong filesystem and network guardrails. Persistence and snapshots would let students resume work with files, code, and execution context intact -supporting longer-running tasks like notebooks and data exploration.”

Cody Little
AI Technical Lead, Department for Education, South Australia



Use Cases for Sandboxes

QR Code
Placeholder
for launch
blog



Agent Workflows

Give AI agents persistent, isolated workspaces that survive across task boundaries



AI Code Execution

Safely run LLM-generated code in isolated environments with instant startup



Platform Building

Build your own platforms on top of the same primitive powering Microsoft services



Burst Workloads

Scale from zero to thousands of sandboxes in response to demand



Secure Multi-Tenant Compute

Strong isolation guarantees for running untrusted workloads from multiple tenants



Interactive User Sessions

Provide each user with their own isolated compute environment

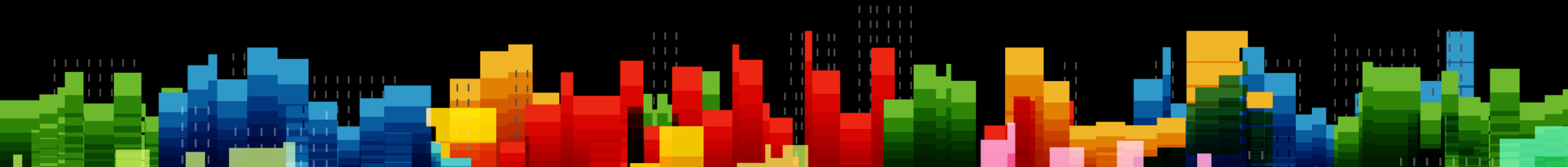
Auger

Customer Highlight



Gopi Prashanth

Chief Scientist - AI & Agents
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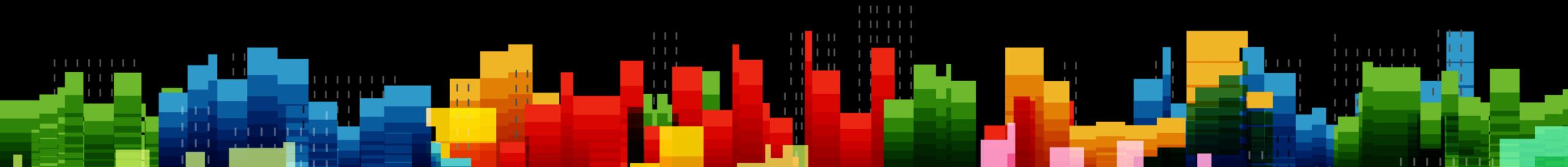
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One Pen. Fifty Actors. Most Are Invisible.

Auger

Raw Mat.	Processing	Components	Assembly	Shipping	Warehouse	Transport	Retail
Oil wells Saudi Arabia	Plastic Resin China	Barrel Molder China	Sub-Assembly Vietnam	Export Port Veracruz	Regional DC Texas	LTL Trucking US Network	Office Depot Retail
Tungsten Mine Vietnam	Carbide Mill Japan	Ball Tip China	Final Assembly Mexico	Customs Mexico Clearance	Amazon FC New Jersey	Amazon Delivery Same Day	Amazon.com Online
Copper Mine Chile	Brass Foundry Taiwan	Ink Cartridge Mexico	Quality Control Mexico	Ocean Freight 30 days	West Coast DC California	B2B Freight Bulk	Walmart Mass Market
Carbon Black Germany	Ink Chemicals India	Clip & Spring Vietnam	ISO Compliance Certification	Air Freight Urgent	Cross-Dock Chicago	Air Express Overnight	Staples Specialty
Natural Gas Qatar	Polymer Plant South Korea	Cap Molder Malaysia	Packaging Line Mexico	Import Port Long Beach	3PL Provider Ohio	Rail Transport Midwest	Consumer End User
Rare Earth China	Solvent Refinery Netherlands	Ink Tube Germany		US Customs Clearance	Cold Storage New York	Last Mile 3PL Urban	Corporate B2B Bulk Buyer
Nickel Mine Indonesia	Steel Wire Mill Germany	Label Printer Vietnam					
50 ACTORS · 8 TIERS · 17 COUNTRIES · MOST ERP SYSTEMS SEE ONLY THE LAST 3							

WHEN A TUNGSTEN MINE IN VIETNAM FLOODS, YOUR ERP DOESN'T KNOW. AUGER HAS ALREADY MITIGATED THE IMPACT.

Auger: The Autonomous Supply Chain

Auger

Models commoditize. The context layer that compounds across every deployment, interaction and decision.

 Built On Ausco	 AI-native	 Context Driven
Domain ontology as the semantic substrate	Agents that share a world model with the business	The right knowledge at the right moment, always



Data Integration: Expensive & Error Prone

Auger

No Reuse. No Scale. Bespoke Every Time.



Data Lives In Silos

ERP, WMS, TMS, Kafka streams, email attachments, all with conflicting representations and schemas.



Unstructured Knowledge Lost

Supplier confirmations, SOPs, expedite requests, passed in email, never integrated into any system.



Bespoke Every Deployment

Each new customer requires months of custom integration work. No reuse, no compounding.



Edge Cases Found Too Late

Fields populated only under certain conditions, system disagreements, aliasing, discovered after go-live.

ADI is Agentic, Iterative & Self-learning.

Auger

1	Understand	2	Map To Ontology	3	Design & Build	4	Deploy
	Profile schema & cardinality AI enriches business meaning Discover joins & FK inference		Apply customer business rules Map entities & functions Mass-balance coherence gate		Architect ETL DAG Generate PySpark on Fabric Customer signs off (HITL)		Push to Microsoft Fabric Run pipelines end-to-end Mass-balance gate + monitor

Foundation



Agent Service



Without World Model: Agents Break.

Auger



No Shared World Model

Every agent builds custom logic. 'Order' means something different to every integration. Nothing composes. Nothing audits consistently.



Schema is Customer-specific

Patterns, KPIs, and learnings from one customer are locked in their schema, they cannot transfer to the next deployment.



Terminology is Fragmented

'DC' vs 'Fulfillment Center'. 'OTW' vs 'In Transit'. The same concept exists under dozens of names, breaking every analysis.

Ausco: World Supply Chain Model

Auger

No Schema Rebuild. One Source Of Truth. Agents Compose Rather Than Rebuild.

 Entities the things	 Actions mutations	 Functions queries + KPIs
SKUs, Locations, Orders, Suppliers, base objects + role instances + plan-vs-actual pairs	Place PO, release shipment, adjust forecast, governed, auditable, reversible write-backs	Days of supply, fill rate, OTIF, defined once, consistent across every agent and screen
 Models optimization + ML	 Workflows multi-step stateful	 Knowledge text + signals
Demand forecast, allocation, N-tier inference, first-class assets any agent can call	S&OP cycles, exception triage, scenario reviews, Temporal-backed durable orchestration	SOPs, contracts, transcripts + live disruption signals, all in the same ontological frame

Without Context, Reasoning is Inconsistent.

Auger

Context Quality is the Performance Multiplier, not the Model.

Without Grounded Context	With The Context Layer
✗ Agent improvises KPI logic, inconsistent answers ✗ No memory of terminology resolutions made before ✗ Context window fills with irrelevant history ✗ Month 1 performance $\approx$ Month 12: Zero learning ✗ New tenants start from scratch every time	✓ AUSCO Functions invoked, same answer, every agent ✓ Validated terminology retrieved from tenant memory ✓ Bounded context packet: only the relevant AUSCO slice ✓ Queries compound, fast path widens with every query ✓ New tenants inherit cross-tenant base from day one

Two systems with the same model and the same data will perform very differently based on context quality.

Auger

Accurate, Grounded & Compounding Context Wins.

Ausco-grounded · Model-agnostic · Compounds With Every Query

Natural Language	→	Context	→	Skills	→	Reason
Query Arrives		Retrieve		Route		Answer
From user, agent, or another product		Multi-hop over AUSCO + STM + LTM + Skills		Fast path (known pattern) or Reasoning Path (discovery)		Response + traceable provenance, grounded

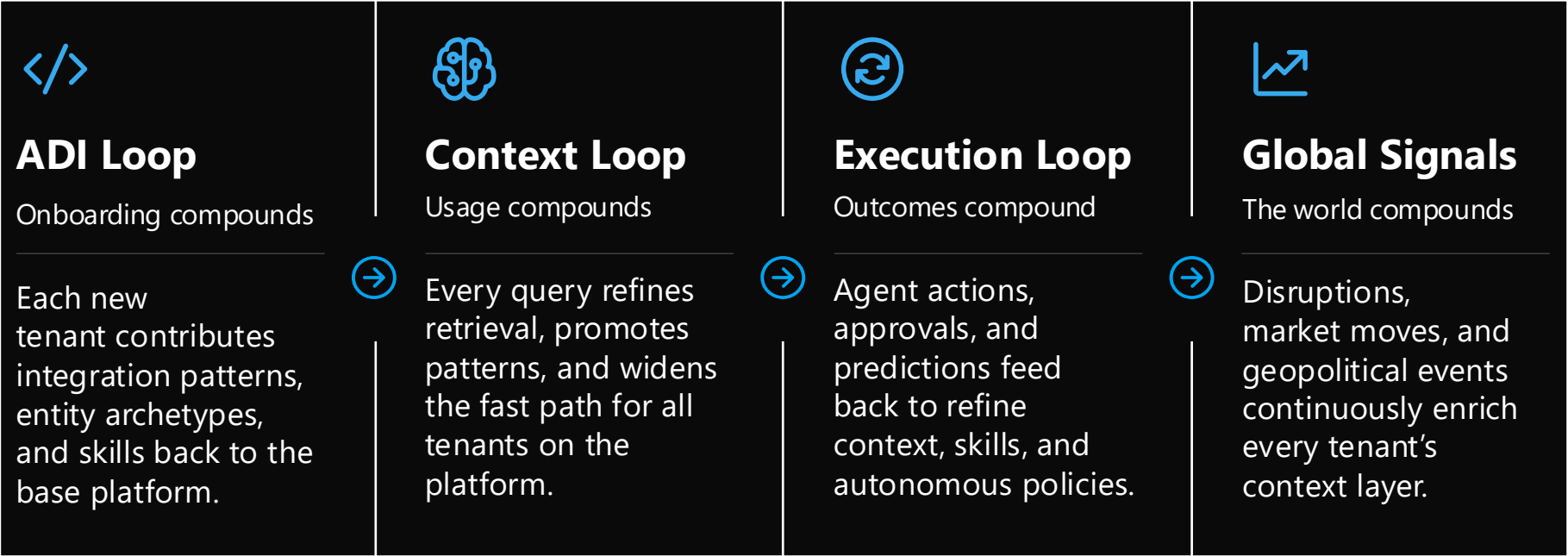
RESOURCES THE CONTEXT LAYER RETRIEVES OVER

Query Arrives	Short-term Memory	Long-term Memory	Skills Library	Global Signals
6 first-class concepts · structural grounding for every query	Working state · turn ledger · resolved entities per session	Validated patterns · KPIs · decision rationales per tenant	Code methodologies graduated from proven real-world usage	Live disruptions, weather, sanctions, market events

Auger: Autonomy That Compounds.

Auger

Four Self-learning Loops, All Converge On AUSCO.



50% Faster executions

85% Autonomous decisions



Key Takeaways

- 1 Azure Container Apps delivers a cohesive agent runtime today – Apps, Sandboxes, Express, and serverless GPUs.
- 2 Azure Container Apps Sandboxes are in public preview – fast, isolated, stateful compute to untrusted code securely by default.
- 3 Proven in production today – GitHub Copilot, Foundry Agent Service, SitecoreAI, EdChat, Auger and more.

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Session Resources



Product Blog

aka.ms/aca/build



Code Sample

[aca/build2026-brk221](https://aka.ms/build2026-brk221)

Get Started Today!



**Azure Container
Apps Sandboxes**

aka.ms/aca/sandboxes



**Azure Container
Apps Express**

aka.ms/aca/express



**Azure Container Apps
Sandboxes Portal**

aka.ms/aca/sandboxes/portal

Explore the session details page
to take action with tutorials,
resources and code

Visit aka.ms/build/evals or scan the QR
code to fill out a session survey

